

Machine Learning for Robotics
(CS4090)

Date: June 9th 2026

Course Instructor

Ms. Sarah Khaleel

Final Exam

Total Time: 3 Hours

Total Marks: 80

Total Questions: 6

88
Section

J. Aiba
Student Signature

- Instruction/Notes:**
- Except for Q1, attempt all questions on answer sheet.
 - Circle the correct option for Q1 once. Choosing multiple options and cutting will result in zero marks for that question.
 - Clearly mention the question number and the respective part number before attempting question. Solve all parts of a specific question in consecutive manner.

CLO 1 (Q1): Demonstrate basic understanding of Machine Learning Algorithms, Neural Networks and optimization methods. **[15 Marks]**

Question 1: Choose the correct option and justify your answer in one line in space with each part. Justification is compulsory for getting marks against correct option. (1*15=15 Marks)

- i) In multiple linear regression, the Normal Equation always finds the global optimum regardless of the learning rate.
- ☒ a. True ☐ b. False
- ii) The Normal Equation has $O(n)$ computational complexity, making it always preferable to gradient descent. True/False?
- a. True ☒ b. False
- iii. Adding more features to a linear regression model always decreases the training MSE.
- ☒ a. True ☐ b. False
- iv. Feature scaling is required for the Normal Equation to work correctly.
- a. True ☒ b. False
- v. If the learning rate α is too small, gradient descent will always converge but may be very slow.
- ☒ a. True ☐ b. False

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- vi. The decision boundary of logistic regression can be a circle or ellipse for 2D data.
☒ a. True ☐ b. False
- vii. Increasing the regularisation parameter lambda in logistic regression always improves test accuracy.
☐ a. True ☒ b. False
- viii. Backpropagation computes gradients in $O(L)$ passes through the network, where L is the number of layers — making it more efficient than computing each gradient separately.
☒ a. True ☐ b. False
- ix. Setting all weights to zero is a good initialisation strategy for neural networks.
☐ a. True ☒ b. False
- x. Dropout during inference (test time) should be applied with the same drop rate as training.
☐ a. True ☒ b. False
- xi. A large value of C in soft-margin SVM leads to a larger margin and more generalization.
☐ a. True ☒ b. False
- xii. K-Means requires the number of clusters K to be specified before running the algorithm.
☒ a. True ☐ b. False
- xiii. The sigmoid activation function is preferred over ReLU in deep hidden layers because it has a smooth, bounded output.
☐ a. True ☒ b. False
- xiv. ReLU activation is strictly better than sigmoid in hidden layers because it is zero-centred.
☐ a. True ☒ b. False
↳ it is better but not because of this reason
- xv. High accuracy on a test set always means the model is good and ready for deployment.
☒ a. True ☐ b. False

CLO 2 (Q2 and Q3): Apply machine learning algorithms such as regression, SVM, clustering, and neural networks to robotics related problems.. [20+20=40 Marks]

Question 2: (5*4= 20 Marks)

1. A binary classifier gives predictions on a test set: TP=50, FP=10, TN=30, FN=10. Compute Accuracy, Precision, Recall, and F1-Score. Which metric matters most if the positive class is "rare cancer case"?
2. Given $\theta = [0.5, 1.0, -0.3]$ and a new sample $x = [1, 2, 3]$ (with bias), compute the prediction. If the true value $y = 3.5$, compute the squared error loss.
3. Find the binary cross-entropy loss for a batch of 3 samples: $y=[1,0,1]$, $h=[0.9, 0.2, 0.4]$. Compute the average loss.
4. For the dataset $x = [1, 2, 3]$ and $y = [1, 3, 2]$, manually compute the MSE cost $J(\theta)$ for:
 - i) $\theta = [0, 1]$ ii) $\theta = [1, 0.5]$, and state which parameter set is better.

Question 3. Perform one full forward and backward pass for a single neuron with sigmoid activation. Given: $w=0.6$, $b=0.1$, $x=2.0$, $y=1$ (true label), learning rate $\alpha=0.5$. Update w and b . (20 marks)

CLO 3 (Q4 to Q6): Analyze robotic learning problems using gradient descent, regularization, activation functions, and computational graphs to improve model performance. [5+10+10=25 Marks]

Question 4. (3+2= 5 Marks)

1. Derive the gradient of MSE cost $J(\theta)$ with respect to θ_j for multiple linear regression. Then write the vectorised gradient descent update rule.
2. If $n = 10,000$, which do you use the Normal Equation or the Gradient Descent and why? Given n features, state the computational complexity of each.

Question 5. A data scientist at a retail company is using K-Means clustering to segment customers based on two behavioural features: average monthly spend (x_1) and visit frequency per month (x_2). After running the algorithm, it converges on two clusters: (1.5+7+1.5=10 marks)

- Cluster 1 - "Casual Shoppers": customers {A, B, C} with centroid $\mu_1 = (1.67, 2.33)$
- Cluster 2 - "Loyal Shoppers": customers {D, E} with centroid $\mu_2 = (4, 6)$

The recorded data for the five sampled customers is:

Customer	A	B	C	D	E
Avg Monthly Spend (x_1)	1	1	3	5	3
Visit Frequency (x_2)	1	2	4	7	5

(a) The data scientist wants to measure how tight the clusters are. Write down the K-Means objective function (WCSS/inertia) in proper mathematical notation that she should minimise.

(b) To validate cluster quality before presenting to the marketing team, she computes how far each customer sits from their cluster's centroid. Calculate the squared Euclidean distance from each customer to their assigned centroid, then compute the WCSS for each cluster and the total WCSS.

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(c) Before settling on $K = 2$, the data scientist had tested multiple values of K . Her notes show the following WCSS values:

K	1	2	3	4	5	6
WCSS	250	100	60	55	52	50

Using the Elbow Method, determine the optimal number of customer segments. Justify your answer by computing the marginal WCSS reduction at each step, and identify where diminishing returns set in.

Question 6: An autonomous robot uses Q-learning to navigate a warehouse.

(10 marks)

Initial Q-table values:

State	Left	Right
S1	(2)	5
S2	4	1

The robot is currently in state S1 and chooses action "Left".

Given: $\alpha=0.4$, $\gamma=0.9$, Immediate reward: $r=-2$, $\max Q(s',a')$ =Maximum future Q-value in next state

After taking action, the robot moves to state S2. Perform the following:

(a) Update the Q-value for: $Q(S1, \text{Left})$

(b) After update, determine which action the robot will prefer in state S1

$$\begin{aligned}
 a &= \\
 \hat{y} &= r + \gamma \times a \\
 Q_{\text{loss}} &= (\hat{y} - a)^2 \\
 Q &= Q + \alpha \times (\hat{y} - Q)
 \end{aligned}$$

- Q1: 15 marks
- ① T
 - ② F
 - ③ T (on training data only)
 - ④ F
 - ⑤ T
 - ⑥ T

- 7) F
- 8) T
- 9) F
- 10) F
- 11) F
- 12) T

- 13) F
- 14) T
- 15) F

Q2: TP=50, FP=10, TN=30, FN=10

5 marks Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} \Rightarrow \frac{50+30}{100} = 0.8 \Rightarrow \boxed{80\%}$

Precision = $\frac{TP}{TP+FP} \Rightarrow \frac{50}{50+10} = 0.833 \Rightarrow \boxed{83.3\%}$

of all total positive how many did we catch? Recall = $\frac{TP}{TP+FN} \Rightarrow \frac{50}{50+10} = 0.833 \Rightarrow \boxed{83.3\%}$

F1-Score = $\frac{2 * (Precision * Recall)}{Precision + Recall} \Rightarrow \frac{2 * 0.833 * 0.833}{0.833 + 0.833}$
 $= \frac{2 * 0.694}{1.66} \Rightarrow \boxed{0.833}$

RECALL is important for rare cancer.

2 Prediction = $h = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$
 $= 0.5(1) + 1(2) + (-0.3)3$

5 marks $\boxed{h = 1.6}$

SE Loss = $(h-y)^2$
 $= (1.6 - 3.5)^2 \Rightarrow (-1.9)^2 \Rightarrow \boxed{3.61}$

3 5 marks 1) $y=1, h=0.9$

Binary cross entropy loss = $L1 = -(y \log(0.9) + (1-y) \log(0.1)) = \boxed{0.1054}$

" $L2 = -(0 \log 0.2 + 1 \times \log(0.8)) \Rightarrow \boxed{0.2231}$

$L3 = -(1 \log(0.4) + 0 \times \log(0.6)) \Rightarrow \boxed{0.9163}$

Average loss = 0.4149

4 5 marks 1) $h(1) = 0 + 1 \times 1 \Rightarrow \boxed{1}$

Error = $1 - 1 = 0$

$h(2) = 0 + 1 \times 2 \Rightarrow \boxed{2}$

Error = $2 - 3 = -1$

$h(3) = 0 + 1 \times 3 \Rightarrow \boxed{3}$

Error = $3 - 2 = 1$

$J = \frac{1}{6} ((0)^2 + (-1)^2 + (1)^2) \Rightarrow \frac{2}{6} \Rightarrow \boxed{0.333}$

$$h(1) = 1 + 0.5(1) \Rightarrow 1.5$$

$$h(2) = 1 + 0.2(2) \Rightarrow 2$$

$$h(3) = 1 + 0.5(3) \Rightarrow 2.5$$

$$J = \frac{1}{6} (0.25 + 1 + 0.25) \Rightarrow \frac{1.5}{6} \Rightarrow \boxed{0.25}$$

$$\text{Error} = 1.5 - 1 \Rightarrow 0.5$$

$$\text{Error} = 2 - 3 \Rightarrow -1$$

$$\text{Error} = 2.5 - 2 \Rightarrow 0.5$$

he (11) ~~that~~ $\theta = [1, 0.5]$ gives better results. As $J \boxed{0.25} < 0.33$

Q3 (20 marks) $w = 0.6, b = 0.1, x = 2, y = 1, \alpha = 0.5$ update w stop

$$z = wx + b$$

$$= 0.6(2) + 1 \Rightarrow \boxed{1.3}$$

(2 marks)

$$a = \sigma(2) = \frac{1}{1 + e^{-2}} \Rightarrow \frac{1}{1 + e^{-1.3}} \Rightarrow \boxed{0.7858}$$

(2 marks)

$$L = -y \log(a) + (1-y) \log(1-a)$$

$$= -[1 \log 0.7858 + 0 \log (0.2142)]$$

$$\boxed{L = 0.2412}$$

(3 marks)

Forward pass

$$\frac{dL}{da} = \frac{-y}{a} + \frac{1-y}{1-a}$$

$$= \frac{-1}{0.7858} + 0 \Rightarrow \boxed{-1.2726}$$

(3 marks)

$$\frac{da}{dz} = a(1-a) \Rightarrow \boxed{0.1684}$$

(3 marks)

$$\frac{dL}{dz} = \frac{dL}{da} * \frac{da}{dz}$$

$$= -1.2726 * 0.1684$$

$$= \boxed{-0.2143}$$

$$\text{OR } \frac{dL}{dz} = a - y$$

$$= 0.7858 - 1$$

$$= \boxed{-0.2142}$$

(simplified)

Backward pass

$$w_{\text{new}} = 0.6 - 0.5(-0.2142)$$

$$b_{\text{new}} = 0.1 - 0.5(-0.2142)$$

$$\Rightarrow \boxed{0.8142}$$

$$\Rightarrow \boxed{0.2071}$$

(2 marks)

(2 marks)

Q4: (1) (3 marks) $J = \frac{1}{2m} * (\sum (x\theta - y))^2$

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} * \sum x_j (x\theta - y)$$

$$\theta = \theta - \alpha \left(\frac{1}{m} \sum (x\theta - y) \cdot x^T \right)$$

OR here

Q4: (2)

(2 marks)

$$n < 10,000$$

→ NE

$$n > 10,000$$

→ GD

For $n = 10,000$:NE (10^{12} operations, too slow)

$$\rightarrow O(n^3)$$

So, GD used, manageable with mini-batching.

$$\rightarrow O(m \times 10,000)$$

$$\Rightarrow O(m \times n)$$

Q5: (a)

(10 marks)

$$J = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2$$

 $C_k \rightarrow$ set of points in cluster k . $\mu_k \rightarrow$ centroidPer cust. sq. distance & WCSS: $\mu_1 = (1.67, 2.33)$

Cluster 1:-

$$A(1,1) = (1 - 1.67)^2 + (1 - 2.33)^2 \Rightarrow 0.449 + 1.769 \Rightarrow 2.218$$

$$B(1,2) = (1 - 1.67)^2 + (2 - 2.33)^2 \Rightarrow 0.449 + 0.109 \Rightarrow 0.558$$

$$C(3,4) = (3 - 1.67)^2 + (4 - 2.33)^2 \Rightarrow 1.769 + 2.789 \Rightarrow 4.558$$

$$WCSS_1 = 2.218 + 0.558 + 4.558 \Rightarrow 7.334$$

Cluster 2:-

$$D(5,7) = (5 - 4)^2 + (7 - 6)^2 \Rightarrow 2$$

$$E(3,5) = (3 - 4)^2 + (5 - 6)^2 \Rightarrow 2$$

$$WCSS_2 = 2 + 2 \Rightarrow 4$$

$$\text{Total WCSS} = 7.334 + 4 \Rightarrow 11.334$$

(1.5 marks)

K = 3 (40 → 5, sharp dropoff)

Q6:- Smaller

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

$$\alpha = 0.4, \gamma = 0.9, r = -2$$

current value: $Q(s_1, \text{left}) = 2$

max future value from s_2 : $\max(4, 1) = 4$

$$Q = 2 + 0.4[-2 + 0.9(4) - 2]$$

$$Q = 1.84$$

(b)

Smaller

updated Q-value for s_1 :-

left { Right
1.84 } 5

Robot prefers: Right